

Figure 4.8: Four-phase Implementation Roadmap with SDLC Stage Mapping

	Pre-thesis (Planning–Design)	Phase I Weeks 1–4	Phase II Weeks 5–8	Phase III Weeks 9–12	Phase IV Weeks 13–16
SDLC Stage Mapping	Planning – Design	Development	Development	Verification – Validation	Maintenance
Primary Focus	Requirements, Architecture, System & Smart Contract Design, Dataset & ML Pipeline	Core Contracts & Infrastructure Development	Lending Lifecycle, Oracle & ML Integration, Front-end & API	Testing, Security, Validation, Simulation, Documentation	Deployment Hardening, Monitoring, Optimization, Handover
Primary Deliverables	• SRS & Use Cases • 4-tier Architecture Design • Smart Contract Spec • DB Schema & API Design • ML Pipeline Design • Project Plan & Risk Register	• Four-tier Core Contracts (WB, NB, LB, Wallet Auth) • Role & Access Control • Wallet Integration • Basic Frontend Shell • DB Schema & Migrations • Unit Tests (Core)	• Lending & Deposit Modules (SavingsVault, FixedDeposit, GroupLendingPool, IBLP, UpwardDeposit) • Oracle Integration • ML Scoring API (RF + IF) • Authority Brief (SHAP) • Frontend Flows (End-to-End)	• Integration & System Tests • Security Testing (Slither, Mythril) • Simulation (300 clients) • Formal Methods (Invariants) • Documentation (Tech + User) • Performance & Gas Analysis	• Testnet Deployment • Monitoring & Alerting • Bug Fixes & Optimization • Backup & Recovery • Final Report & Handover • Future Work Roadmap
Optional Agent (dashed)	—	—	—	Conversational Agent (MCP) – Prototype Development	Agent Integration & Evaluation
Key Milestone/ Outcome	Design Complete (Spec Baseline)	Core Platform Operational	End-to-End Loan Flow with ML Scoring	Validated & Tested System (MVT Ready)	Deployed, Monitored & Handover Complete

